

while Loop in C

Theory:

The 'while' loop is used to repeatedly execute a block of code as long as a given condition is true.

Steps:

1. Initialize the variable
2. Check the condition
3. Execute the loop body
4. Update the variable

Syntax:

```
while(condition) {  
    // statements  
}
```

Full Example:

```
#include <stdio.h>  
  
int main() {  
    int i = 1;  
    while(i <= 5) {  
        printf("%d ", i);  
        i++;  
    }  
    return 0;  
}
```

do-while Loop in C

Theory:

The 'do-while' loop is similar to 'while', but the loop body is executed at least once before the condition is checked.

Steps:

1. Execute the block
2. Check the condition
3. Repeat if condition is true

Syntax:

```
do {  
    // statements  
} while(condition);
```

Full Example:

```
#include <stdio.h>  
  
int main() {  
    int i = 1;  
    do {  
        printf("%d ", i);  
        i++;  
    } while(i <= 5);  
    return 0;  
}
```

Functions in C

Theory:

A function is a block of code that performs a specific task. Functions help in code reusability and modular programming.

Steps:

1. Function Declaration
2. Function Definition
3. Function Call

Syntax:

```
returnType functionName(parameters) {  
    // code block  
}
```

Full Example:

```
#include <stdio.h>  
  
int add(int a, int b) {  
    return a + b;  
}  
  
int main() {  
    int result = add(10, 20);  
    printf("Sum = %d", result);  
    return 0;  
}
```

Default (Library) Functions in C

Theory:

C provides built-in library functions like printf, scanf, strlen, sqrt, etc. These functions are defined in header files.

Common Examples:

1. printf() - prints output
2. scanf() - reads input
3. strlen() - string length
4. sqrt() - square root

Syntax:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
#include <string.h>
```

Full Example:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char str[] = "Hello";
```

```
    printf("Length = %lu\n", strlen(str));
```

```
    printf("Square Root of 16 = %.2f\n", sqrt(16));
```

```
    return 0;
```

```
}
```

User-defined Functions in C

Theory:

User-defined functions are created by the programmer to organize code into logical blocks.

Steps:

1. Declare the function
2. Define the function
3. Call the function from main()

Syntax:

```
void functionName() {  
    // code block  
}
```

Full Example:

```
#include <stdio.h>  
  
void greet() {  
    printf("Welcome to C Programming!\n");  
}  
  
int main() {  
    greet();  
    return 0;  
}
```

